Monday, January 13, 2025 1:03 PM

vector space :- V, binary operation $+$, scalar multiplication.

Subspace :- A subset $\phi \neq W$, is called subspace if

① $w_1, w_2 \in W \Rightarrow w_1 + w_2 \in W$

② $w \in W$, $\alpha w \in W$ $\forall\, \alpha \in F$, $w \in W$.

* $M_n(\mathbb{R})$ is vector space over $\mathbb{R}$

$\hookrightarrow$ $n \times n$ matrices with entries from $\mathbb{R}$.

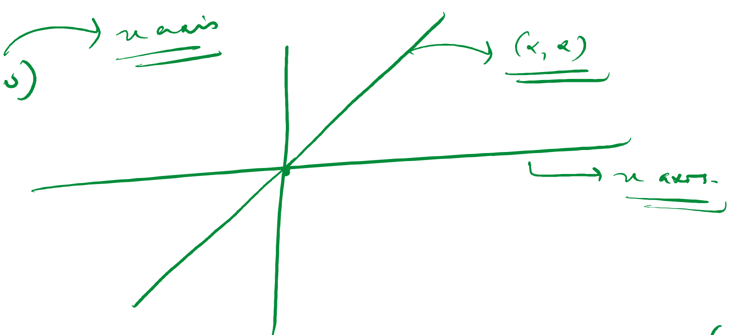
* $D_n(\mathbb{R}) \subseteq M_n(\mathbb{R})$ (Subspace)

$\hookrightarrow$ set of diagonal matrices

Linear Combination of vectors :-

$\{v_1 \in V\}$ L.C of $v_1 = \{\alpha v_1 \mid \alpha \in F\}$.

$\mathbb{R}^2 \longrightarrow (1,0) \longrightarrow$ x axis



$\alpha(1,0)$
$\Rightarrow (\alpha, 0)$

$(0,1) \longrightarrow$ y axis

$(1,1)$

$(0,0)$
$\alpha(0,0) = $

$\{v_1, v_2 \mid v_1, v_2 \in V\} \xrightarrow{\text{L.C.}} \{\alpha v_1 + \beta v_2 \mid \alpha, \beta \in F,\ v_1, v_2 \in V\}$

$\mathbb{R}^2$ $\mathbb{R}$ $\{(1,0), (0,1)\} \xrightarrow{\text{L.C.}} \boxed{\{\alpha(1,0) + \beta(0,1) \mid \alpha, \beta \in \mathbb{R}\}} \subseteq \mathbb{R}^2$

$\parallel$
$\mathbb{R}^2$

$V \rightarrow$ vector space $\longrightarrow$ $\alpha v_1, \beta v_2 \in V \rightarrow \alpha v_1 + \beta_2 v_2 \in V$

$v_1, v_2 \in V$

$\alpha v_1 + \beta v_2 \in V$, $\alpha, \beta \in F$

$\mathbb{R}^2$

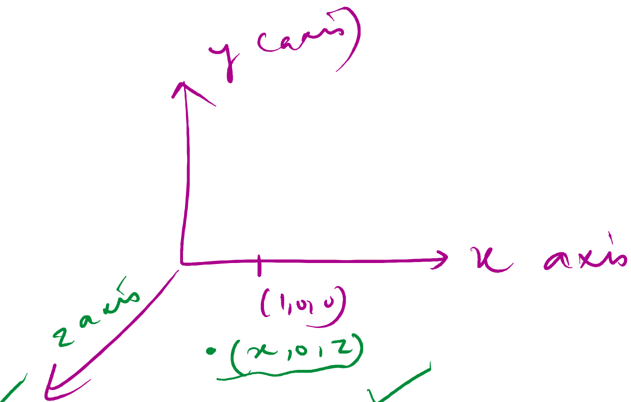
$$(x,y) \mid x \in \mathbb{R},\ y \in \mathbb{R}$$

$$(x,y) = x(1,0) + y(0,1)$$

$\mathbb{R}^3$

$$(1,0,0)$$

$$\alpha(1,0,0) \to (\alpha,0,0)$$



L.C. of $\boxed{\{(1,0,0),(0,1,0)\}} \to \alpha(1,0,0) + \beta(0,1,0)$, $\alpha,\beta \in \mathbb{R}$

L.C. of $\{(1,0,0),(0,1,0),(0,0,1)\} \to \alpha(1,0,0) + \beta(0,1,0) + \gamma(0,0,1)$

$\boxed{\mathbb{R}^3}$

Linear Combination of vectors

Let V be a vector space over field F. Let $S = \{v_1, v_2, \ldots, v_k\}$ be a subset of V. Then linear combination of vectors in S is

$$L.C(S) = \{\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_k v_k \mid \alpha_i \in F,\ 1 \le i \le k\}.$$

Example: $V = M_2(\mathbb{R})$

$$S = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$$

$L.C.(S) =$ Diagonal matrices.

Claim: $L.C(S)$ is also subspace of V.

$S = \{v_1, \ldots, v_k\}$

Claim: L.C(S) is also subspace of V.

$W_1, W_2 \in L.C(S)$ $\qquad S = \{v_1, \dots, v_k\}$

$w_1 = \alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_k v_k$

$w_2 = \beta_1 v_1 + \beta_2 v_2 + \dots + \beta_k v_k$

$w_1 + w_2 = (\alpha_1+\beta_1) v_1 + (\alpha_2+\beta_2) v_2 + \dots + (\alpha_k+\beta_k) v_k \in L.C(S)$

$\alpha_i, \beta_i \in F = \boxed{\mathbb{R} \text{ or } \mathbb{C}}$

$\alpha_i + \beta_i \in F$

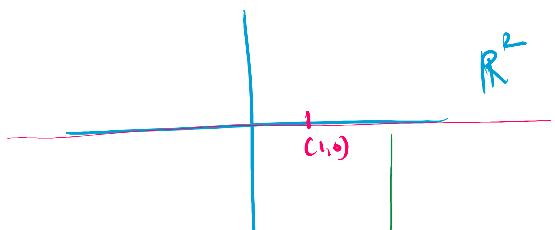
$r w_1 = r\alpha_1 v_1 + r\alpha_2 v_2 + \dots + r\alpha_k v_k$ $\qquad \alpha_i \in F$, $r\alpha_i \in F$, $r \in F$.

Claim: If $S = \{v_1, v_2, \dots, v_k\}$ is subset of V. Then the smallest subspace of V containing S is L.C(S).

$\mathbb{R}^2$

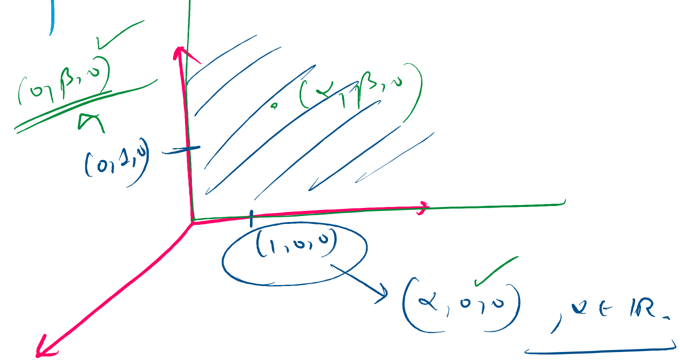
$S = \{(1,0)\}$

$L.C(S) = x\text{-axis}$



$\mathbb{R}^3$ $\qquad S = \{(1,0,0), (0,1,0)\}$

$L.C(S) = xy$ plane



Linear span / generators.

Let V be a vector space over field F. Let $S = \{v_1, v_2, \dots, v_k\}$ be a subset of V. The subspace

$$V = W = \{\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_k v_k \mid \alpha_i \in F, 1 \le i \le k\}$$

is called linear span of S $\qquad$ vector space V

$V = W$ ...

is called linear span of S

* $\{v_1, v_2, \dots, v_k\}$ are called generators of subspace W of vector space V.

$\mathbb{R}^2$

- x axis is linear span of $(1,0)$.
- $(1,0)$ is generators of x axis.

$\mathbb{R}^2$

- $\mathbb{R}^2$ is linear span of $\{(1,0), (0,1)\}$
- $\boxed{(1,0) \text{ \& } (0,1) \text{ are generators of } \mathbb{R}^2}$

- $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ are generators of subspace of Diagonal matrices $D_2(\mathbb{R})$.

- Are generators of a vector space unique?

$(x,y) \in \mathbb{R}^2$

$(x,y) = \frac{x}{2}(2,0) + \frac{y}{2}(0,2)$.

- Is $\{(1,0), (1,1)\}$ a generating/spanning set of $\mathbb{R}^2$??

$(x,y) = (x-y)(1,0) + y(1,1)$.

$\alpha(1,0) + \beta(1,1) = (0,0)$

$(\alpha+\beta, \beta) = (0,0)$

$\beta = 0$

$\alpha + \beta = 0$

$\alpha = 0$

$\{(1,0), (0,1), (1,1)\}$

$(x,y) = x(1,0) + y(0,1) + 0(1,1)$

$= (x-y)(1,0) + 0(0,1) + y(1,1)$

$= (x - y/2)(1,0) + \frac{y}{2}(0,1) + \frac{y}{2}(1,1)$

$\boxed{-2(1,0) + 1(2,0) = (0,0)}$

$-2, 1$

$\boxed{\{(1,0), (2,0)\} \quad \mathbb{R}^2}$

linearly dependent

$(1,1)$

$(1,0) \quad (2,0)$

- Let V be a vector space over F. Two vectors $\boxed{\{v_1, v_2\}}$ S are called linearly dependent if $v_1 = \alpha v_2$ for some $\alpha \in F$.

$v_1 \in L.C.(S \setminus \{v_1\})$ $\{v_2\}$ $\leftarrow$ $v_1 \in L.C.(v_2)$

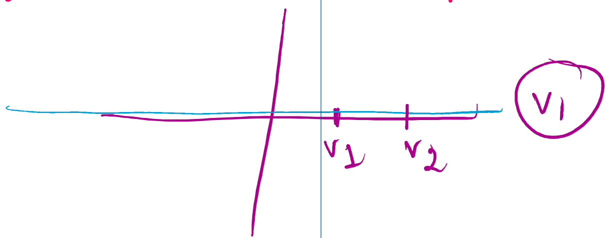
- Let V be a vector space over F. A subset $\{v_1, v_2, \dots, v_k\} = S$

called linearly dependent if $\exists \; v_j \in S$ such that

- Let V be a vector space over F. ... S is called **linearly dependent** if $\exists\ v_j \in S$ such that

$v_j \in$ linear combination of $S \setminus \{v_j\}$.

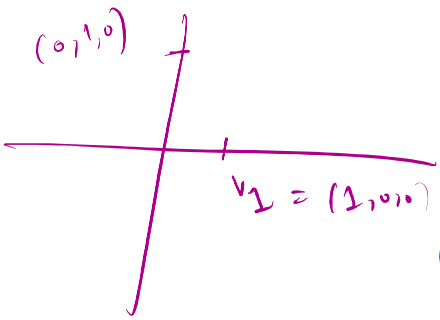
$\mathbb{R}^2$ $\{(1,0), (0,2)\}$



L.C. of v_1

$\{v_1, v_2\}$ is L.D

$v_2 = \alpha v_1$



$\{(1,0,0), (0,1,0)\}, (2,3,0)\}$ L.D??

$-2(1,0,0) + (-3)(0,1,0) + 1(2,3,0) = 0$

- A subset $S \subset V$ is called linearly independent if it is NOT linearly dependent.

- Let V be vector space over F. A subset $S = \{v_1, v_2, \dots, v_k\}$ is linearly dependent in V, if $\exists$ non-zero scalars $\alpha_1, \alpha_2, \dots, \alpha_k \in F$ such that

$$\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_k v_k = (0, \dots, 0)$$ zero vector

- A subset 'S' is linearly independent in V if $\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_k v_k = 0$ ✓

$$\Rightarrow \alpha_1 = 0, \alpha_2 = 0, \dots, \alpha_k = 0$$

$\{(1,0), (1,1)\} \subseteq \mathbb{R}^2$ Linearly independent.

$\alpha(1,0) + \beta(1,1) = (0,0)$

$\alpha = 0, \beta = 0$

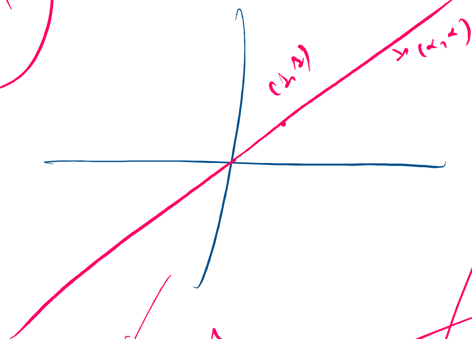
v_1 $(0,2)$

$\{(1,1), (2,5)\} \subseteq \mathbb{R}^2$

$\{(1,1), (2,5)\} \subseteq \mathbb{R}^2$

$\mathbb{R}^n$

$\mathbb{R}$



$\alpha_1(1,1) + \alpha_2(2,5) = (0,0)$

$(\alpha_1 + 2\alpha_2, \alpha_1 + 5\alpha_2) = (0,0)$

$\alpha_1 + 2\alpha_2 = 0$

$\alpha_1 + 5\alpha_2 = 0$

$-3\alpha_2 = 0$

$\Rightarrow \alpha_2 = 0$

$\Leftarrow \alpha_1 = 0$

$\{(1,1), (2,5)\}$ is linearly independent